

UNIT 03

ORGANIC COMPOUNDS



ORGANIC CHEMISTRY:

"All those compounds which contain carbon as an essential element along with hydrogen, oxygen, nitrogen, sulphur, phosphorus, halogens, etc. are called organic compounds."

OR

"Organic chemistry is that branch of chemistry which deals with the study of compounds of carbon and hydrogen (hydrocarbons) and their derivatives."

Exception: However, there are several compounds like carbon monoxide (CO), carbon dioxide (CO₂), carbon disulphide (CS₂), carbonates (CO₃²⁻), bicarbonates (HCO₃⁻), Cyanides (CN⁻), thiocyanates (SCN⁻) etc are studied in inorganic chemistry because of their properties.

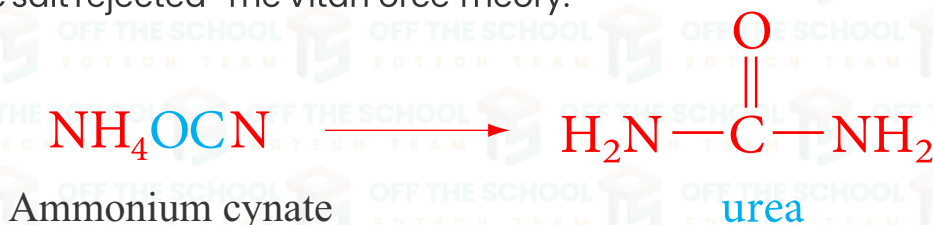
VITAL FORCE THEORY:

"Organic substances can only be obtained from living sources (animals & plants) & cannot be synthesized in laboratory."

- Organic compounds were produced from plants & animals under a mysterious type of force, which is called as vital force.
- Synthesized organic matter in laboratory by ordinary methods.
- Organic compounds can only be isolated from organic sources, cannot from inorganic sources.
- Example:** Urea was isolated from animal waste & rubber was isolated from rubber plant.

OBJECTION OF VITAL FORCE THEORY:

In 1828, a chemist FREDRICH WOHLER accidentally discovered Urea by heating together Ammonium Sulphate & Potassium Cyanate. Both are inorganic salts. Previously Urea was extracted from animal urine but formation of urea from an inorganic salt rejected "The Vital Force Theory."

**SOURCES OF ORGANIC COMPOUNDS:**

- Organic compounds are abundant in nature.
- They are found in various natural sources such as fossil fuel, animal, plants & micro-organism

FOSSILS REMAINS:

The remnants of animal & plants that have been preserved in nature.

- When living organism such as animals' dies or plants shatters it gets buried by sediments without being decomposed. Therefore, organic compounds remain within the entity.
- The buried remains get compressed with the passage of time & organic compounds present in remains slowly transfer to earth crust as fossil fuel.
- Fossils are non-renewable form of energy.
- Fossil fuel includes:

Sources of Organic compounds

1. Natural gas

2. Coal

3. Petroleum

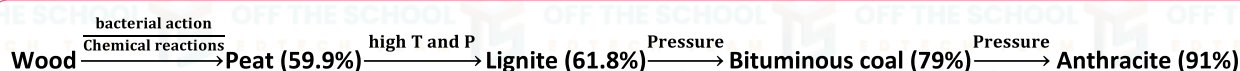
NATURAL GAS:

- Natural gas is one of the important sources of obtaining simple organic compounds.
- It was produced by the decomposition of marine organism millions of years back.
- It is a mixture of methane, ethane, propane & butane with 85–95% methane.
- It is regarded as best fuel as its combustion causes less environmental pollution.
- The composition of Sui Gas is:

Methane 94.60%	Butane 0.17%
Propane 0.28%	Ethane 4.05%

COAL:

- Coal is derived from plants present under the earth crust.
- Plants buried million years ago under high temperature & pressure converts to coal.
- Various types of coals are found, which are as under:



- Various organic compounds can be extracted from coal.
- Coal can be gasified or liquefied to produce wide range of organic compounds such as methane, ethene, ethyne, hexane etc.

PETROLEUM:

- Petroleum known as “rock oil” or “crude oil” is a viscous liquid & a rich source of many organic compounds.
- It is the complex combination of alkanes, cycloalkanes & aromatic hydrocarbons.
- It was produced by the decomposition of marine organism millions of years back.
- It is used for power generation & transportation fuel.
- Fractional distillation is the process of separation of components of a liquid mixture on the basis of the difference in their boiling points in a fractionating column.
- The products extracted from petroleum through fractional distillation are: oil, diesel, naphtha & paraffin wax. Gasoline, kerosene.

DESTRUCTIVE DISTILLATION:

The process in which coal is heated in the absence of air to produce wide range of useful organic products is called destructive distillation.

- The process required high temperature about 400 – 900°C. The process is carried in closed system.
- Four chief products are obtained.

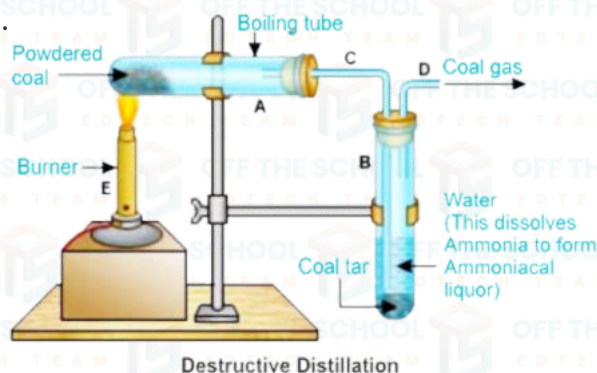
1. Coal: it is greyish-black hard solid with 98–99% carbon. Used as fuel & reducing agent in industry.

2. Coal tar: it is thick black liquid, through fractional distillation various important compounds are obtained such as toluene, xylene, benzene, naphthalene & phenol etc.

3. Coal Gas: it is the mixture of methane & water gas. It is highly inflammable used in heating & municipal lighting.



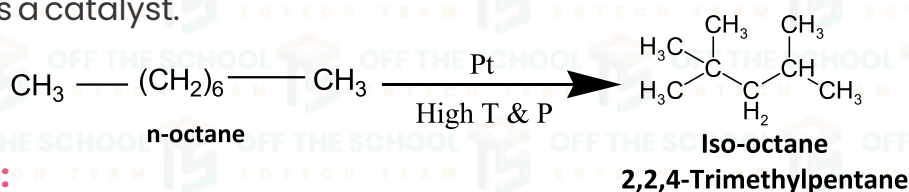
4. Ammonia Liquor: it is the mixture of ammonium compounds & liquid ammonia from bituminous coal.



REFORMING OF PETROLEUM:

“The process of conversion of gasoline of straight chain alkanes to form branched chain alkanes in order to improve the octane rating of fuel in internal combustion engine.”

The process is carried out at high temperature & pressure in the presence of platinum as a catalyst.



KNOCKING:

OCTANE NUMBER OR RATING:

- In order to judge the quality of petrol there is number (depends upon arrangement of atoms C in chain) that defines fuel ability to resist knocking during combustion.
- The quality is determined by research octane number (RON).
- Straight chain alkanes have low octane number example, RON of n-octane = 0.
- Branched chain alkanes such as Iso-octane has high RON = 100.

ADDING KNOCK INHIBITOR:

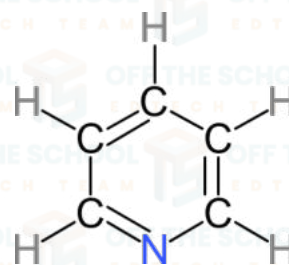
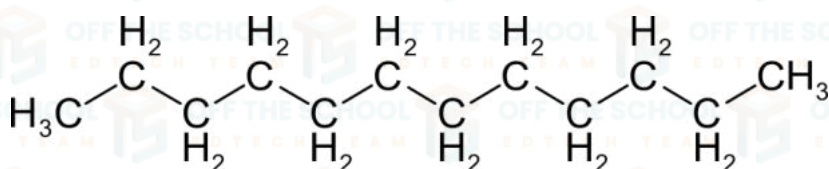
- Knock Inhibitor is the chemical which when added to the fuel, helps in reduction of knocking in internal combustion engine as it moderates the explosion in cylinder of engine.
- Previously, tetraethyl lead (TEL) $\text{Pb} (\text{C}_2\text{H}_5)_4$ was extensively used as a knock inhibitor. But, it was found to be responsible for environmental pollution.

CATENATION:

The natural ability of a carbon atom to covalently bond itself (with other carbon atom) forming long chain, branched chain & ring together.

Carbon has highest catenation ability because:

1. It is tetravalent & it can co-ordinate (link) from all side with 4 four-carbon atoms.
2. The small size is also favors catenation because bonded electrons remains closer to nucleus hence attraction & stability is high. Let's have a look dodecane & pyridine:



FUNCTIONAL GROUP:

The active part of an organic compounds which give the unique properties to the compounds.

Important Functional Groups in Organic Compounds

Functional Group	Compound Type	Prefix or Suffix of Name	Lewis Structure	Ball-and-Stick Model	Systematic Name (Common Name)
$\text{C}=\text{C}$	alkene	-ene			ethene (ethylene)
$\text{—C}\equiv\text{C—}$	alkyne	-yne	$\text{H—C}\equiv\text{C—H}$		ethyne (acetylene)
$\text{—C—}\ddot{\text{O}}\text{—H}$	alcohol	-ol			methanol (methyl alcohol)
$\text{—C—}\ddot{\text{X}}\text{:}$ (X = halogen)	haloalkane	halo-			chloromethane (methyl chloride)
$\text{—C—}\ddot{\text{N}}\text{—}$	amine	-amine			ethanamine (ethylamine)
$\text{O}=\text{C—H}$	aldehyde	-al			ethanal (acetaldehyde)
$\text{O}=\text{C—C—C—H}$	ketone	-one			2-propanone (acetone)
$\text{O}=\text{C—}\ddot{\text{O}}\text{—H}$	carboxylic acid	-oic acid			ethanoic acid (acetic acid)
$\text{O}=\text{C—}\ddot{\text{O}}\text{—C—H}$	ester	-oate			methyl ethanoate (methyl acetate)
$\text{O}=\text{C—}\ddot{\text{N}}\text{—H}$	amide	-amide			ethanamide (acetamide)
$\text{—C}\equiv\text{N:}$	nitrile	-nitrile			ethanenitrile (acetonitrile, methyl cyanide)

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HOMOLOGOUS SERIES:

The members of same class of compounds arrange ascending order of their molecular weight with a common difference of methylene group (CH_2) or by 14 a.m.u. in molecular mass.

The members of the homologous series are called homologue.



Characteristics of Homologous Series:

- All members are prepared by same general method.
- Members are differentiated by methylene group or by molecular weight of 14 a.m.u.
- Physical properties gradually vary with increasing molecular weight.
- All members of series are similar in chemical properties.
- Homologous series is represented by a general formula.

C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}	$C_nH_{2n+1}X$	$C_nH_{2n+1}OH$	$C_nH_{2n+1}NH_2$
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Properties of homologue:

- Melting & boiling point increases with increasing number of carbon atoms. Solubility decreases with increasing number of carbon atoms.
- At ordinary temperature & pressure C1-4 are gases, solids. C5-7 are liquids, & higher members are solids.

Names and Formulas of the First Ten Unbranched Alkanes

Name	Number of carbons	Molecular formula	Structural formula
methane	1	CH_4	CH_4
ethane	2	C_2H_6	CH_3CH_3
propane	3	C_3H_8	$CH_3CH_2CH_3$
butane	4	C_4H_{10}	$CH_3CH_2CH_2CH_3$
pentane	5	C_5H_{12}	$CH_3(CH_2)_3CH_3$
hexane	6	C_6H_{14}	$CH_3(CH_2)_4CH_3$
heptane	7	C_7H_{16}	$CH_3(CH_2)_5CH_3$
octane	8	C_8H_{18}	$CH_3(CH_2)_6CH_3$
nonane	9	C_9H_{20}	$CH_3(CH_2)_7CH_3$
decane	10	$C_{10}H_{22}$	$CH_3(CH_2)_8CH_3$

CHARACTERISTICS OF ORGANIC COMPOUNDS:

Carbon-Based Structure: The backbone of these compounds is formed by carbon atoms.

Covalent Bonding: They primarily involve covalent bonding.

Diversity: Exist in a vast number of varieties due to the versatility of carbon to form chains, rings, and branches.

Types of Organic Compounds:

Hydrocarbons: Compounds consisting of only hydrogen and carbon.

Examples: Methane, Ethane.

Functional Groups: Oxygen, nitrogen, or other atoms attached to the carbon chain, imparting different properties. Examples: Alcohols, Ketones, Amines.

Physical Properties:

State: Can be gases, liquids, or solids.

Solubility: Generally insoluble in water but soluble in organic solvents.

Boiling and Melting Points: Vary widely; influenced by molecular weight and structure.

Chemical Properties:

Combustion: Most organic compounds burn in air or oxygen to give CO_2 , water, and heat.

Reactivity: Varied reactivity depending on functional groups present.

Isomerism in Organic Compounds:

Structural Isomerism: Compounds with the same molecular formula but different structural arrangements.

Stereoisomerism: Same molecular formula but different spatial arrangement of atoms.

Importance in Daily Life:

Basis of Life: Key components of all life forms (DNA, proteins, carbohydrates, etc.).

Applications: Extensively used in industries like pharmaceuticals, plastics, fuels, and dyes.

Organic Synthesis:

Laboratory Preparation: Methods to synthesize organic compounds in a lab setting.

Industrial Production: Large-scale synthesis for commercial purposes.

Previous Year's Questions



Homologous series:

1. Define Homologous series with examples. (2025, Supp.16, 2016, S.14, 14, S.12, 10, 09,07,04,)
2. Write a short note on Homologous series? (1998)
3. What are Homologous series? Write down the formulae & names of first five members of a homologous series? (2006)

Functional group:

1. Define Functional group. (2025,17,15,Supp.14,2009,05,04)
2. Define the Functional group. Name the functional groups in the following compounds: (i) $\text{C}_2\text{H}_5\text{COCl}$ (ii) CH_3CONH_2 (2015)
3. Write a short note on functional group. (2000)

General Topics:

1. Define Catenation and octane number. (2025,17,15,12,07,03,Supp.16,14,10)
2. Write the classification of organic compounds. (2015)
3. Define Reforming of petroleum? (2025,2014)
4. Briefly describe the natural sources of organic compounds. (2013,02)

